Nonlinear Modeling of Cycling

Christina Cavalluzzo

May 2025

1. Introduction

A half Ironman consists of a 1.2 mile swim, a 56 mile bike, and a 13.1 mile run. Cycling is the middle and longest leg of a half Ironman. During the bike, the cyclist wants to go as fast as possible while also preserving their legs for the half marathon run that follows. The goal of this project is to model and simulate cycling using nonlinear modeling of the mechanics of the bike and the biomechanics of the cyclist, looking at different aerodynamic postures, hoping to maximize speed while preserving energy for the run.

2. Mathematical Cycling Modeling

The total resistive forces acting on a cyclist can be summed to a total resistive force to be

$$F_{total} = F_{roll} + F_{slope} + F_{accel} + F_{aero}$$

where $F_{roll} = mgC_r$, $F_{slope} = mgsin(\theta)$, $F_{accel} = ma$, and $F_{aero} = \frac{1}{2}\rho C_d A V^2$. m is the weight of the cyclist plus the weight of the bike, g is the gravitational constant 9.81, C_r is the coefficient of rolling resistance, θ is the slope of the terrain, a is the acceleration of the bike, ρ us the density of air, C_d is the coefficient of rolling resistance, A is the drag frontal area, and V is the velocity of the bike. Knowing the relation between the velocity, V, of the bike, the total resistance forces, and the power required to propel the bike at a velocity past all the resistive forces of $P_{total} = F_{total} * V$, we have our total mechanical power model for the bike to be the following.

$$P_{total} = mgC_rV + mgsin(\theta)V + maV + \frac{1}{2}\rho C_d A V^3$$

It is important to note that the aerodynamic drag force provides the most resistance to a cyclist due to the velocity being cubed.

At a given muscle contraction velocity, the muscle can produce a certain amount of force. Hill's equation, $(v + b)(F + a) = b(F_0 + a)$, shows a hyperbolic force-velocity relationship of muscle contraction. Using the relation $P_{muscle} = F(v) * v$, we can rearrange to get a power muscle equation, shown here:

$$P_{muscle}(v) = \left(\frac{b(F_0 + a)}{v + b} - a\right) * v$$

where v is the muscle contraction velocity, is the maximum isometric force, and a and b are Hill's constants.

These two equations can be coupled to link the muscle power to the cyclist's power demand. Assuming an idealized model and ignoring drivetrain losses, the cyclist's acceleration is the muscle's power minus the restrictive forces shown here:

$$m\dot{V} = \frac{1}{r}\left(\frac{b(F_0 + a)}{v + b} - a\right) - \left[mgC_r + mgsin(\theta) + ma + \frac{1}{2}\rho C_d A V^2\right]$$

3. Assumptions and Parameters

To model and look at more specifics in this problem, specific assumptions were made and parameters calculated.


Figure 1


Figure 2


Figure 3


Figure 4

There are three main bike positions on a road bike. There is the hands on the hoods, being the most "normal" and relaxed bike position and providing the most amount of area affecting drag, shown in figure 1. Then another position is the hands in the drop bars, which is when the biker is slightly crouched on the handle bars, giving a smaller frontal area affecting drag, shown in figure 2. Finally, there is the aero bar position, shown in figure 3, where the cyclist is extended reducing the frontal area by a lot. Figure 4 shows a fourth cycling position, a plank, where the cyclist reduces the frontal area by almost a third, however this cycling position is illegal in most races. From these and other images, the approximate areas can be estimated by comparing known areas in the image. These areas are 0.363 m^2 for the hands on hoods, 0.343 m^2 for hands in drop bars, 0.323 m^2 for hands in the aero bars, and 0.118 m^2 for the plank position.

The weight of the cyclist plus the bike was calculated from the experimented system (Christina and her bike), and is 69.4 kg. The coefficient of rolling resistance, Cr, is 0.003, which is typical for a road bike. The density of air, r, is known to be 1.2 kg/m^3 at sea level. The drag coefficient, Cd, is 0.9 which is typical for road bikes. It is assumed that the cyclist will always be in a gear that produces optimal cadence.

	Hoods (A=0.363)	Drop Bars (A=0.343)	Aero Bars (A=0.323)	Plank (A=0.118)
Flat (0°)	23.49 mph	23.94 mph	24.38 mph	33.78 mph

Uphill (+5°)	8.72 mph	8.75 mph	8.83 mph	-
Downhill (-5°)	42.50 mph	43.62 mph	44.74 mph	71.58 mph

Table 1: steady state velocities at equilibrium points

Table 1 shows the steady state velocities at the equilibrium points for flat ground, a 5° uphill, and a 5° downhill, for each of the 4 cycling positions. As seen, at slower speeds like going uphill, the aerodynamic drag does not matter as much for speed, because all the speeds for the different cycling positions are within 0.11 mph. However, aerodynamics are much more significant at faster speeds like downhill, where the difference between hands on the hoods vs hands in the aero bars is a 2.24 mph difference. For a long biking race, this will make a major difference in time. The plank position, reaching 71.58 mph on a downhill, shows just how important it is to reduce aerodynamic drag at high speeds, but it also helps explain why a plank is illegal in races, and is also an unsafe position.

4. Closed Loop Stability of a PI-Regulated Model

To evaluate the stability of the speed-tracking control system, the closed-loop dynamics were modeled using a PI-controller. The state space model has the primary state variables to be $x_1 = V - V_{ref}$, which is the speed deviation, and $x_2 = \int (V_{ref} - V) dt$, which is the integral error. This gives the state space equations to be

$$\dot{x_1} = \frac{P_{mus}(t) - P_{tot}(x_1 + V_{ref}, i)}{m(x_1 + V_{ref})}$$

$$\dot{x_2} = -x_1$$

In this model, the total resistive power required to maintain speed on the bike is approximated to be $P_{tot}(V) = \alpha V + \beta V^3$. α accounts for the rolling and gravitational resistances and β accounts for the aerodynamic drag. The controller input $u(t)$ is the normalized muscle activation level between 0 and 1 where 1 is the maximum effort.

This system is linearized about the operating point $V = V_{ref}$. Taking the partial derivatives yields

$$\frac{\partial f}{\partial V} = -\frac{\alpha + 3\beta V_{ref}^2}{mV_{ref}}$$

$$\frac{\partial f}{\partial u} = -\frac{1}{mV_{ref}}$$

Defining $a = -\frac{\alpha + 3\beta V_{ref}^2}{mV_{ref}} < 0$ and $b = \frac{1}{mV_{ref}} > 0$ and substituting these constants into the state equations, the simplified equations are

$$\dot{x_1} = ax_1 + b\tilde{u}$$

$$\dot{x_2} = -x_1$$

where $\tilde{u} = u - u^*$ is the small deviation from the equilibrium input u^*. The PI controller is

$$u(t) = u^* - K_p x_1 + K_i x_2$$

And using $\tilde{u}$ it becomes $\tilde{u} = -K_p x_1 + K_i x_2$. Substituting this into the linearized dynamics gives the closed-loop system,

$$\dot{x_1} = (a - bK_p)x_1 + bK_i x_2$$

$$\dot{x_2} = -x_1$$

These equations can be written in matrix form as

$$\dot{x} = A_{CL} x$$

where $A_{CL} = \begin{pmatrix} a - bK_p & bK_i \\ -1 & 0 \end{pmatrix}$. The characteristic equation of this closed-loop system is

$$\lambda^2 - (a - bK_p)\lambda + bK_i = 0$$

Using the Routh-Hurwitz stability criterion for a second-order polynomial $s^2 + a_1 s + a_0$, this can be compared to the cyclist's characteristic equation to get the parameters $a_1 = -(a - bK_p)$ and $a_0 = bK_i$. This system is locally asymptotically stable if $a_1 > 0$ and $a_0 > 0$, which provides the condition

$$K_p > \frac{a}{b} \text{ and } K_i > 0$$

Since $a < 0$ and $b > 0$, any positive PI gains, $K_p > 0$ and $K_i > 0$, will satisfy the condition ensuring asymptotic stability of the closed-loop system around the cycling posture's equilibrium.

5. Bifurcations

This project has strong real-world applications, and analyzing bifurcations helps connect the mathematical results to meaningful qualitative insights about cyclist performance.

5.1 Varying Slope

First, varying the slope was analyzed. Taking the equilibrium equation, $P_{mus} = mgC_{rr}V + mgsin(\theta)V + \frac{1}{2}\rho C_d A V^3$, where the rider's available power equals the total resistive power requirement, it can be rewritten where γ_1 replaces the roll and slope resistances and γ_2 replaces the aerodynamic drag. This gives this equation:

$$P_{mus} = \gamma_1(\theta)V + \gamma_2 V^3$$

where γ_1 is being analyzed. As θ increases, γ_1 increases. At the bifurcation, $\theta = \theta_{crit}$, the discriminant of the cubic velocity becomes zero, $\frac{dP_{tot}}{dV} = 0$. This is solved to get $\gamma_1 + 3\gamma_2 V^2 = 0$, and solve to get $V^2 = \frac{-\gamma_1}{3\gamma_3}$. To get real velocity, the right hand side is less than or equal to 0, so there is a bifurcation at $\gamma_1 = 0$ and real possible solutions disappear. This gives the following.

$$\theta_c = -\arcsin(C_{rr})$$

This is a saddle node bifurcation. This means that at θ_{crit}, the equilibrium disappears and the cyclist can no longer maintain motion, or at a certain point the hill is too steep to bike up.

5.2 Varying Drag Area

Varying the drag area follows a very similar process as varying the slope, however instead of varying γ_1, γ_2, where $\gamma_2 = \frac{1}{2}\rho C_d A$, is being changed. Once looking at the bifurcation and finding a critical area,

$$A_c = -\frac{2mg(C_{rr} + \sin(\theta))}{3\rho C_d V^2}$$

it is concluded that this is also a saddle-node bifurcation. This means that as A approaches A_c, the drag becomes too large to maintain any real solution for V, and the cyclist stalls.

5.3 Varying Integral Gain

To analyze the effect of varying the integral gain, K_i, the linearized closed loop system around the cycling equilibrium is started with. The state space model of this system is the following.

$$\dot{x_1} = (a - bK_p)x_1 + bK_i x_2$$

$$\dot{x_2} = -x_1$$

The characteristic equation for the eigenvalues, λ, is given by

$$\lambda^2 - (a - bK_p)\lambda + bK_i = 0$$

This quadratic equation determines the system's stability and oscillatory behavior as K_i changes. A Hopf bifurcation occurs when the eigenvalues become purely imaginary, such as $\lambda = \pm jw$. This happens when $a - bK_p = 0$ and $bK_i > 0$, so, $K_p = \frac{a}{b}$ and $K_i > 0$. This creates a critical gain at $K_p = \frac{a}{b}$ where a $= \frac{\frac{dP_{tot}}{dV}}{mV_{ref}}$. At this Hopf bifurcation, the eigenvalues are

$$\lambda = \pm j\sqrt{bK_i}$$

which indicates the system enters a limit cycle regime. This analysis is important when avoiding unwanted oscillations and ensuring accurate speed tracking.

5.4 Varying Force Limit

Lastly, the force limit, F_0, was varied. Taking the maximum muscle power output, $P_{mus} \leq F_0 V$, this can be inserted into the equation $P_{tot} = \gamma_1 V + \gamma_2 V^3$, then solving for V^2 gives $V^2 = (F_0 - \gamma_1)/\gamma_2$. If $F_0 < \gamma_1$, then $V^2 < 0$ which means there is no forward equilibrium. Therefore, this is also a saddle-node bifurcation, where the critical bifurcation is the following.

$$F_0^{crit} = \gamma_1 = mg(C_{rr} + \sin\theta)$$

This means that if the power output is capped below the terrain's demand, the cyclist's motion stalls.

6. Implementing a Controller

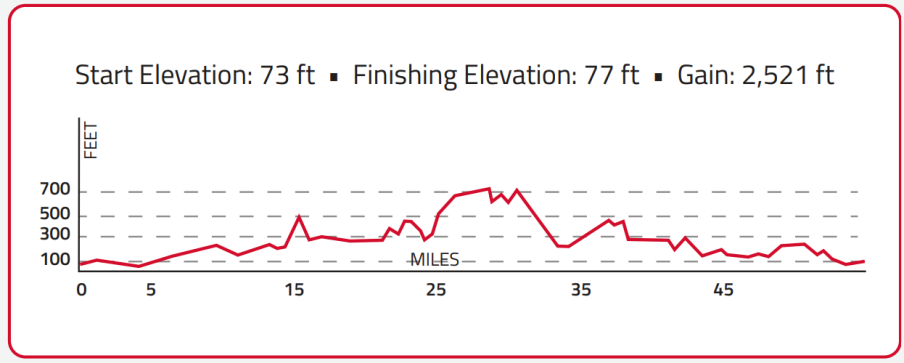
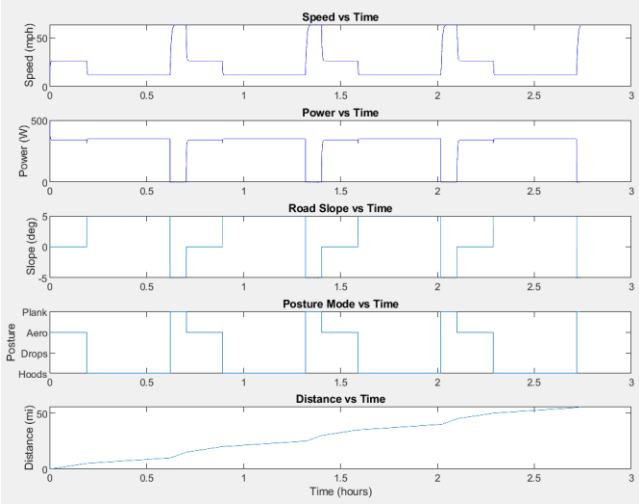


Figure 5

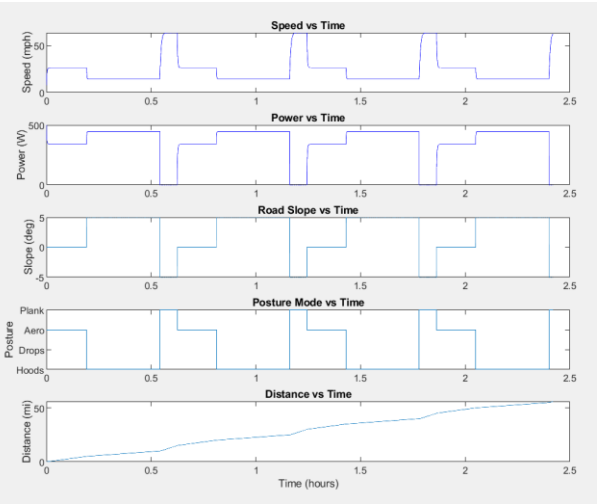
A controller can be implemented into Matlab and simulated to test this project's theories. The parameters of this simulation include two courses that are each 56 miles long, and the cyclist starts from rest and does not stop the entire ride. Two terrains are tested. The first terrain is a repeating synthetic course that cycles through 0°, +5°, -5° inclines every 5 miles. The second terrain is the terrain of the Western Massachusetts Ironman 70.3, estimated using the terrain map shown in figure 5. Two controllers are tested on each of the terrains. The first controller is modeling steady pacing at a steady effort. α, the controller, is the effort from 0 to 1 where 1 is the most cycling effort a cyclist could produce. The strategy with this controller is to have the initial α to be 1, then settle into a steady effort where α = 0.8. The second controller has dynamic adaptive pacing that changes the power based on the slope and speed of the cyclist. For this controller, the uphills have $\alpha = 1$, the downhills allow for the cyclist to coast at $\alpha = 0$, if the downhill slope is < 2°, there is light pedaling at $\alpha = 0.5$, and when the terrain is flat $\alpha = 0.8$. Along with these changes in power, the cyclist also changes their cycling position based on the slope of the terrain. If the slope is > 3°, the cyclist is on the hoods. If the slope is between 1° and

3°, the cyclist is on the drop bars. If the slope is less than 1°, the cyclist is in the aero bars. This is all shown in the Matlab code and can be seen in the supplemental materials.




```
>> simulation_trial1
Controller: steady, Course: repeating
Total time to complete 56 miles: 2 h 44 min 16.0 s
Average speed: 20.45 mph (9.14 m/s)
Average power output (while pedaling): 345.5 W
```

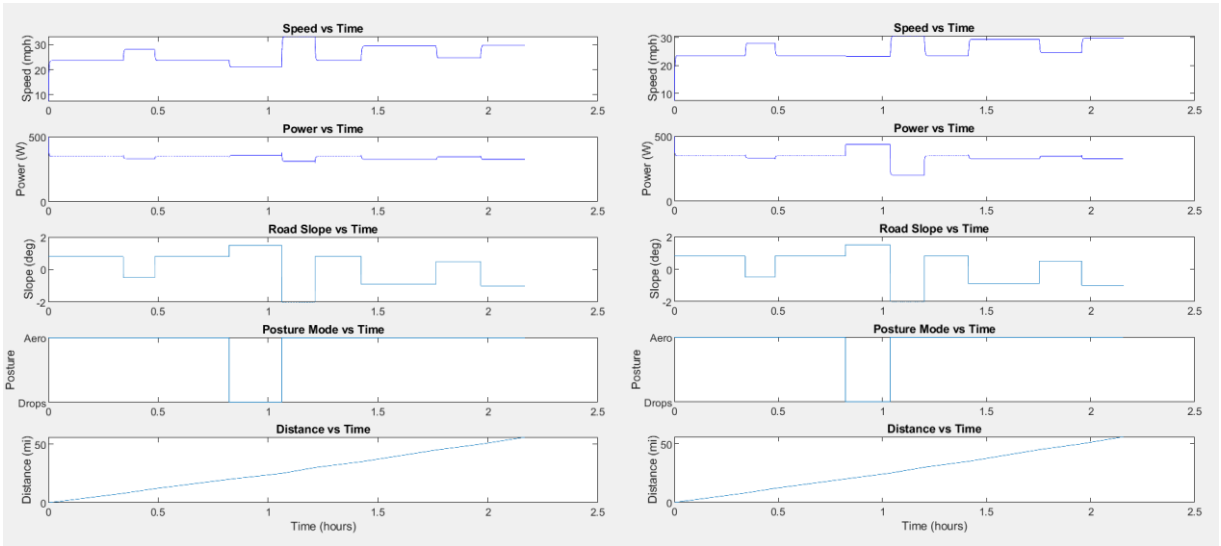
Figure 6




```
>> simulation_trial1
Controller: adaptive, Course: repeating
Total time to complete 56 miles: 2 h 25 min 11.0 s
Average speed: 23.14 mph (10.34 m/s)
Average power output (while pedaling): 408.9 W
```

Figure 7

Figure 6 shows the repeating hills with the steady controller, and figure 7 shows the repeating hills with the adaptive controller. When comparing metrics, it is seen that the adaptive controller produces a lower time to complete the course by almost 20 minutes, and produces a faster average speed by almost 3 mph.



```
>> simulation_trial1
Controller: steady, Course: ironman
Total time to complete 56 miles: 2 h 10 min 9.0 s
Average speed: 25.81 mph (11.54 m/s)
Average power output (while pedaling): 340.1 W
```

Figure 8

```
>> simulation_trial1
Controller: adaptive, Course: ironman
Total time to complete 56 miles: 2 h 9 min 30.0 s
Average speed: 25.94 mph (11.60 m/s)
Average power output (while pedaling): 339.6 W
```

Figure 9

Figure 8 shows the Western Massachusetts half Ironman with the steady controller, and figure 9 shows the same terrain with the adaptive controller. When comparing these metrics, it is seen that the adaptive controller produces a lower time to complete the course by about 1 minute and produces a small faster average speed. The average power output required by the cyclist also drops with the adaptive controller. These simulations show that an adaptive strategy is optimal for performance.

7. Conclusion

This project modeled cycling dynamics and then nonlinearly analyzed them. Through these nonlinear mathematics, applications, and experimentation, it is concluded that it is best to use aero bars whenever at the highest speeds to maximize cycling speed and minimize cycling time. It was also found that in a hilly terrain, it is best to have adaptive pacing and modulated power output, and in the real world, the cyclist is their own controller. Therefore, it is strategic to work the uphills and cruise on the downhills.

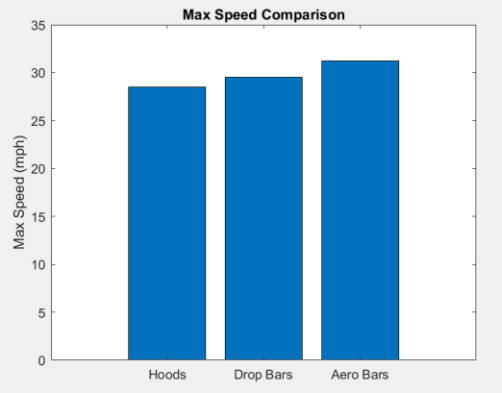


Figure 10

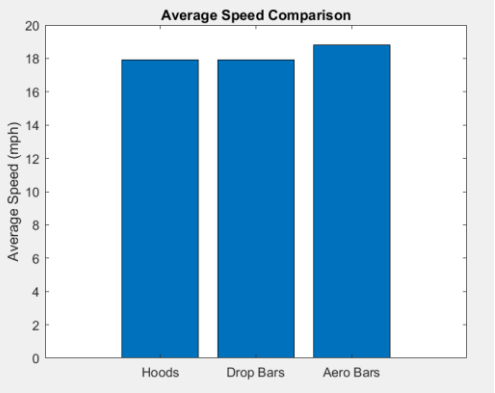


Figure 11

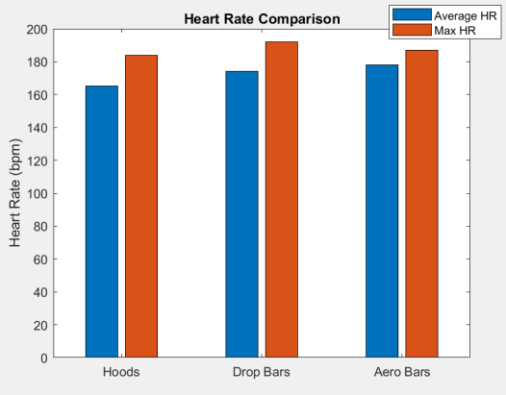


Figure 12

This project was tested by the author in real-world conditions. She chose a 2.5 mile loop that had some hills and no stops. She biked on this path three times, each in a different cycling position (hoods, drop bars, and aero bars). She wore a Garmin vivoactive 4S sports watch to track and capture data on the rides. It was found that the max speed and the average speeds were higher for less aerodynamic frontal area shown in figure 10 and 11 respectively, which matches the modeling in this project. Figure 12 shows Christina's heartrate comparison for the 3 cycling positions, which they have a slight increase when frontal area increases. This perhaps makes sense because when more relaxed in a hood position, the heartrate would be lower. However, some argue the aero position is more relaxed, so this is an area of future research.

Another area of future research is to try to minimize an athlete's VO2 max, which is also related to heartrate. With a lower VO2 max, the trend is that athletes have stronger endurance, so this is a path of future nonlinear modeling. This project can also be taken further to incorporate different wind and environmental conditions, to see how those affect cycling. Finally, future research could include training modeling, to optimize training to then optimize races.

8. References

[1] Course Overview. (2025). IRONMAN. https://www.ironman.com/races/im703-western-massachusetts/course

[2] Dahmen, T. (n.d.). Computing a Field of Optimal Pacing Strategies for Cycling Time Trials Introduction, Related Work, and Target Application. https://www.uni-konstanz.de/mmsp/pubsys/publishedFiles/Da13a.pdf

[3] H.K. Khalil. (2002). Nonlinear Systems (3rd ed). Upper Saddle River, NJ: Prentice Hall.

[4] Muelaner, J. (1998, August 7). Experiments Prove Equation for Bicycle Power. Better Bicycles. https://betterbicycles.org/science/speed/validation-of-bicycle-power-equation/

[5] Muscle Tension-Velocity. (n.d.). Www.brown.edu. https://www.brown.edu/Departments/Engineering/Courses/En123/Lectures/HillEqn.htm

[6] Rinard, D., & Allen, J. (n.d.). Bicycle Drag Force Formulas (R. Pivit, Ed.). Www.sheldonbrown.com; Klara Agil. https://www.sheldonbrown.com/rinard/aero/formulas.html